Solvency Field Theory V13.10

What a Clock Is Actually Doing

Time as the count of completed history

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This paper does not derive time. It defines it, and then shows that one definition accounts for eight facts about time that are ordinarily explained separately or not at all — using no free parameters and no new machinery. What that definition costs, and what would falsify it, are stated at the end.

1 Every equation has a t in it

Not one of them says what t is.

We can measure time to eighteen decimal places. We can predict how much of it will pass on a satellite, in a basement, on a moving train, near a collapsing star, and we are right every time to the limit of the instruments. We built a civilisation on being right about it.

And if you ask a physicist what time *is*, the honest answers are: a coordinate, a parameter, a dimension, the thing entropy increases along. Every one of those says how to *use* it. None says what the thing is.

Relativity comes closest. It says your elapsed time is the length of your path through spacetime — proper time, the arc length along your worldline. That is a formula, and it is correct, and it tells you nothing about what is happening. A length is a measurement of something. What is being measured?

Quantum mechanics does worse, and does it in a way that should be more famous than it is. Every observable quantity in quantum mechanics has a mathematical operator attached to it. Position has one. Momentum, energy, angular momentum, spin — all of them. Time does not. Pauli proved that it cannot: a proper time operator is incompatible with energy being bounded below, which it is. So time sits inside every quantum equation as a parameter that the theory itself cannot treat as a property of anything.

That is not a small gap. It is the most-used quantity in physics being the one thing the theory cannot describe.

2 Six facts, and nothing that explains them together

Before proposing anything, here is what we actually know. All of it is measured.

Clocks disagree, and the disagreement is real. In 1971, Hafele and Keating flew caesium clocks around the world on commercial aircraft. The eastward clocks came back 59 nanoseconds behind the ones that stayed home. The westward clocks came back ahead. This is not a subtlety of instrumentation. Your GPS receiver corrects for it continuously or it would put you in the wrong city by nightfall.

How much time passes depends on the route. Two clocks that separate and reunite generally show different totals, and which one shows more depends on the paths, not on who was "really" moving.

Light has no elapsed time at all. Not a very small amount. Zero, exactly, along any null path.

Time has a direction and the fundamental laws do not. Nearly every microscopic law works equally well run backwards. The world does not.

Every observable has an operator except this one. As above.

And every object carries its own. There is no master clock anywhere. There is no experiment that finds one. There is no "now" that spans the universe, and relativity is emphatic that looking for one is a category error.

Six facts. Take any standard account and you will find it explains some of them well, one or two by hand, and at least one not at all.

3 The proposal

Here it is, and it is one sentence.

> **You do not move through time. Time is the count of what you have completed.**

Not a stage you cross. Not a river that carries you. A tally.

Everything that persists does so by doing something over and over — in this framework, a closed process that returns to itself and writes an irreversible record each time it does. Your elapsed time is the number of those completions. Not a measurement of duration. *The duration.*

The rest of this paper is what follows from that, and it follows without adding anything.

The muon, which makes it concrete in one paragraph

A muon is a heavy electron that comes apart. Left alone it lasts about 2.197 microseconds, and in that span it completes 5.613×10^{16} internal cycles before it fails. That count is the same every time.

Cosmic rays manufacture muons about fifteen kilometres up, moving at roughly 0.998 times the speed of light. At that speed, in 2.197 microseconds, a muon covers 657 metres. It should die nine miles above your head.

Detectors at sea level catch them constantly. About one per square centimetre per minute is passing through you right now.

The usual telling is that time ran slowly for the muon. Here is the other telling:

> The muon did less. It got through exactly the 5.6×10^{16} cycles it always gets through. Nothing slowed down and nothing was stretched. It simply took longer *by our bookkeeping* for it to finish the same amount of doing.

The count never changed. Only our accounting of it did. That is the whole idea, and the rest is consequences.

4 Eight things that follow

1. Why light has no time. A photon never returns to itself. It has no closed process, so nothing accumulates. Its elapsed time is not small; there is no tally to read. This is structural, not a limit of a formula. Whatever a photon's existence is like from the inside — and the phrase may not mean anything — it is not brief. It has no duration at all.

2. Why matter does. A massive particle is a process caught in a loop. It returns, and each return is one more completed event. That is the only difference, and it is the difference between having a clock and not having one.

3. Why heavier means finer, not faster. The tick rate is set by the mass: $\nu_0 = mc^2/h$. An electron completes 1.24×10^{20} cycles per second; a proton completes 1836 times as many. But they age identically. A proton does not live faster than an electron — it has a *finer clock*. A stopwatch reading milliseconds does not run faster than one reading seconds. Elapsed time is the count divided by the natural rate, $\tau = N/\nu_0$, and everything in your body agrees on that even though the raw counts differ by three orders of magnitude.

4. Why the two kinds of time dilation are the same thing. In relativity, motion slows clocks and gravity slows clocks, and the two effects are computed differently and taught in different chapters. Here they are one effect. There is a single speed budget. Spending it on getting somewhere leaves less for returning to yourself; sitting deep in a gravitational well means neighbouring records are already occupying the space your next write needs, so completions are deferred. Either way, fewer returns per unit of somebody else's bookkeeping. Nothing is slowed. The *return* is delayed.

5. Why time has a direction. Completed records cannot be overwritten. That is not a statistical tendency that could in principle reverse if the molecules cooperated — it is the rule that makes a record a record. The arrow is not emergent here. It is the axiom, and everything else emerges from it.

6. Why every object carries its own. Nothing needs to consult anything. Your tally is yours; a distant galaxy's is its own. There is no master clock because there is nothing for one to be a master *of*. This is the part that sounds most radical and is in fact the least controversial — relativity has said there is no universal present for over a century. What is added here is a reason: a count is a local fact about a local process, and local facts do not require a global register to be true.

7. Why time stops at a horizon. At a black hole's edge, the deferral fraction reaches one half and the dilation factor goes to zero. In this language the write queue is full. No new record can be laid down, so no new time is made. "Time stops" is usually delivered as a paradox. Here it is a bookkeeping statement: the ledger is at capacity.

8. Why there is no time operator. An operator represents a property of a system in its current state. A count of completed events is not a property of a state — it is a *history*. You cannot read it off the thing as it is now, because it is not in the thing as it is now; it is the record of what the thing has done. Histories do not have operators, and the anomaly that has sat unexplained since Pauli becomes the expected result.

Eight consequences. One rule. No adjustable numbers anywhere.

5 A clock does not measure time. It makes it.

This is the sentence that separates the proposal from a rephrasing, so it gets its own section.

In the ordinary picture, a clock is an instrument. Somewhere out there a quantity called time is elapsing, and a good clock estimates it accurately while a bad clock estimates it poorly. Clocks can be wrong.

Here, there is nothing to estimate. The count *is* the duration. Which means something strange and, I think, clarifying:

> A clock cannot be inaccurate about its own elapsed time.

It can be broken. It can be badly built, badly read, or disagree wildly with the clock next to it. But it is not making an estimate of an external quantity that it might be getting wrong, any more than a tally of coins in a jar might be inaccurate about how many coins are in the jar. Two clocks that disagree are not one right and one wrong. They did different amounts.

This also disposes of a question that has confused people for a hundred years. When the travelling twin returns younger, what *happened* to the missing time? Nothing happened to it. There was never a shared quantity being divided up. One of them completed fewer events than the other. There is no missing time in the same sense that there is no missing distance when two hikers take different routes to the same summit and one of them walked further.

6 The scale of the thing

Some numbers, because the picture is easier to hold when it has sizes.

	duration	in electron ticks
One electron cycle	8.09×10^{-21} s	1
Fastest interval yet measured (2020)	2.47×10^{-19} s	31
One heartbeat	0.85 s	1.05×10^{20}
Flying east around the world	-59 ns	-7.29×10^{12}
A human life	80 years	3.12×10^{29}
The age of the universe	13.8 Gyr	5.4×10^{37}

Two of those deserve a second look.

A heartbeat is a hundred billion billion electron cycles. Whatever you feel as a moment, your constituents got through 10^{20} completed events inside it.

And flying east around the world costs you seven trillion cycles relative to staying home. That is not a thought experiment. Hafele and Keating measured it with caesium clocks and a commercial airline ticket in 1971, and the answer matched prediction. You can buy that experiment.

7 What it costs

An honest proposal states its bill.

There is no universal now. Nothing anywhere is happening "at the same time" as anything else in any absolute sense. Relativity already said this; here it is unavoidable rather than merely true, because there is no global thing for events to be simultaneous *in*.

There is no flow. Time does not pass or move or run. Those words describe something changing with respect to time, and time cannot change with respect to itself. What actually happens is that records accumulate. The sensation of flow is what accumulation feels like from inside an accumulating thing.

The present is not special. It is simply the leading edge of your own record — the most recent write. It feels unique because you are standing on it.

And your total is not a judgement. Two travellers reunite with different tallies. Neither is behind. Neither lost anything. They did different amounts, and both records are complete.

8 Where this sits

This is not the first proposal that time is not fundamental, and saying so is not a weakness — having neighbours means the idea is in a landscape rather than in a vacuum.

Causal set theory takes spacetime to be discrete and holds that order plus number equals geometry; its time is the accumulation of spacetime elements. Rovelli's relational quantum mechanics and thermal time hypothesis make time relative to a state rather than fundamental. Barbour removes it entirely and keeps only configurations. Page and Wootters derive it from a system's entanglement with a clock subsystem.

What is proposed here shares the family resemblance — discrete, accumulated, relational, not fundamental — and differs on one specific point:

> The counting is done by *objects*, and the rate is set by the object's *mass*: $\nu_0 = mc^2/h$.

Causal sets count elements of spacetime. Page–Wootters makes the clock a subsystem alongside the thing being timed. None of them ties the tick rate to the mass of the thing doing the ticking. That identification — mass *is* clock rate, so mass and duration are one fact counted twice — is what makes this a different proposal rather than a restatement of an existing one.

9 What would show this is wrong

If a heavier particle ages faster rather than merely finer. The claim is that a proton's clock has 1836 times the resolution of an electron's and identical elapsed time. If any experiment shows a heavier species genuinely ageing more per second rather than resolving more finely, the proposal fails immediately.

If proper time turns out to be continuous where it should be grainy. If time is a count, there is a smallest interval for any given species: h/mc^2, which for an electron is 8.09×10^{-21} seconds. The fastest interval ever measured — a photoionisation delay in molecular hydrogen, reported in 2020 — is 2.47×10^{-19} seconds. That is a factor of thirty-one away. This is not a comfortable margin, and it is not a margin that requires new physics to close; it requires better attosecond sources, which are improving.

That is unusual for a foundational proposal about time. Most of them are safe forever. This one has an experimental frontier approaching it from two orders of magnitude out.

And if a photon is ever found to have accumulated anything, the first consequence in §4 is wrong and the rest follows it down.

10 Claim boundary

This paper does not derive time. It proposes a definition and shows what follows from it.

It does not derive quantum mechanics, quantum gravity, or the value of any constant. It does not explain why there is an arrow rather than no arrow — it takes the irreversibility of completed records as a starting rule, and is explicit that this is where the arrow enters rather than where it is explained.

What it claims is this. One definition, stated in a sentence and carrying no adjustable parameters, accounts for the null result for photons, the existence of a clock for matter, the resolution-versus-rate distinction across masses, the unification of the two dilations, the arrow, the absence of a universal present, the horizon, and the missing operator.

Eight things that are usually eight separate explanations, or in the last case no explanation at all.

Whether that is the right definition is not settled here. That it is a definition worth testing — and that it can be tested, from two orders of magnitude away, with instruments that exist — is the claim.

Status. V13.10 narrative statement. Definitional proposal with derived consequences; no new constants, no retuning of prior results. Companion derivation canon to follow.

Originator and lead author: Shaun Fosmark. Narrative: Claude/Opus. Derivation and audit canon: GPT-5.6 SOL. Prior material drawn from V10.6, V11.4, V11.5, V13.1-V3 and V13.2.